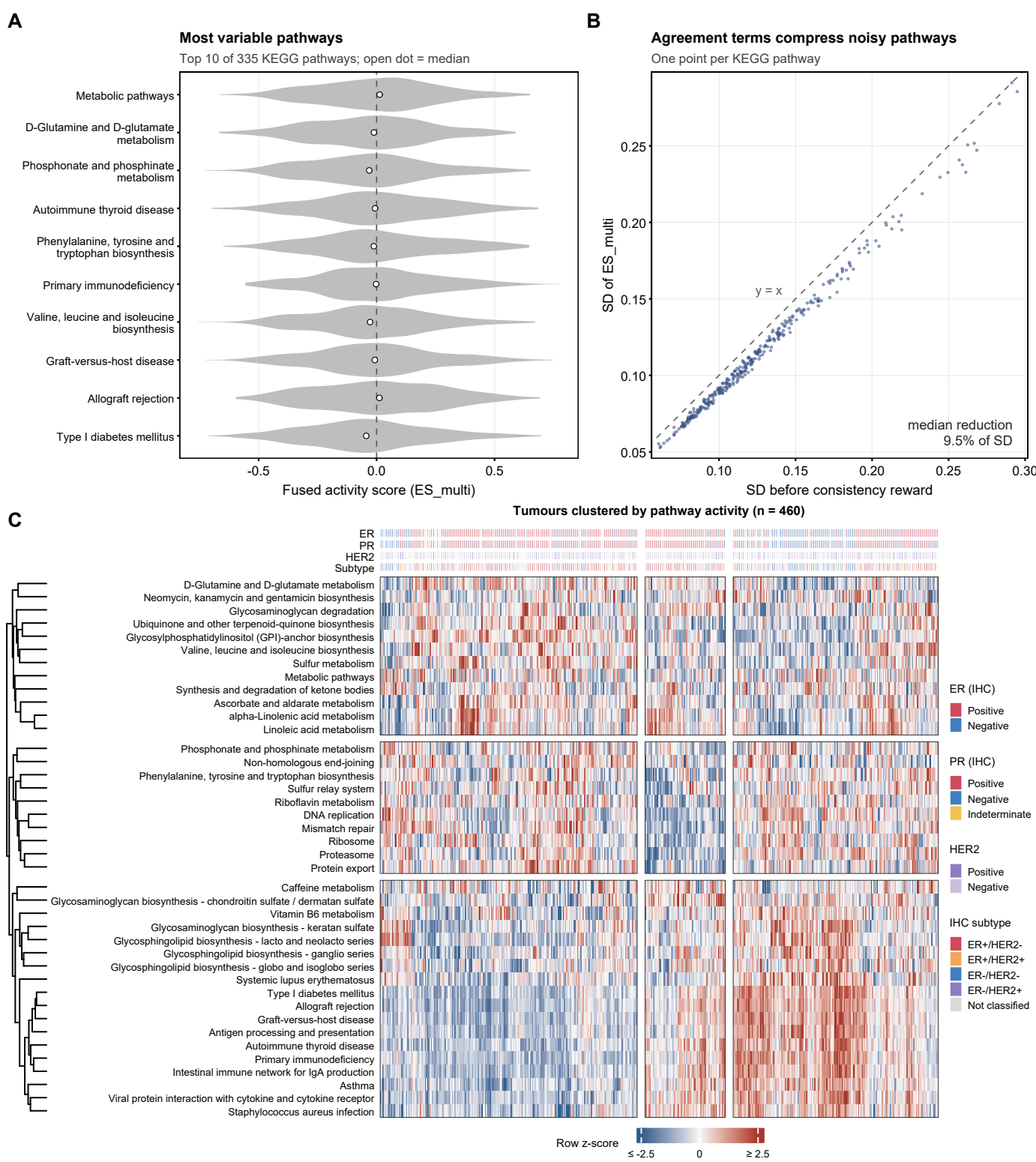

A
Most variable pathways
Top 10 of 335 KEGG pathways; open dot = median
Metabolic pathways
D-Glutamine and D-glutamate metabolism
Phosphonate and phosphinate metabolism
Autoimmune thyroid disease
Phenylalanine, tyrosine and tryptophan biosynthesis
Primary immunodeficiency
Valine, leucine and isoleucine biosynthesis
Graft-versus-host disease
Allograft rejection
Type I diabetes mellitus
-0.5
0.0
0.5
Fused activity score (ES_multi)
B
Agreement terms compress noisy pathways
One point per KEGG pathway
SD of ES_multi
SD before consistency reward
y = x
median reduction
9.5% of SD
0.05
0.10
0.15
0.20
0.25
0.30
C
Tumours clustered by pathway activity (n = 460)
ER
PR
HER2
Subtype
D-Glutamine and D-glutamate metabolism
Neomycin, kanamycin and gentamicin biosynthesis
Glycosaminoglycan degradation
Ubiquinone and other terpenoid-quinone biosynthesis
Glycosylphosphatidylinositol (GPI)-anchor biosynthesis
Valine, leucine and isoleucine biosynthesis
Sulfur metabolism
Metabolic pathways
Synthesis and degradation of ketone bodies
Ascorbate and aldarate metabolism
alpha-Linolenic acid metabolism
Linoleic acid metabolism
Phosphonate and phosphinate metabolism
Non-homologous end-joining
Phenylalanine, tyrosine and tryptophan biosynthesis
Sulfur relay system
Riboflavin metabolism
DNA replication
Mismatch repair
Ribosome
Proteasome
Protein export
Caffeine metabolism
Glycosaminoglycan biosynthesis - chondroitin sulfate / dermatan sulfate
Vitamin B6 metabolism
Glycosaminoglycan biosynthesis - keratan sulfate
Glycosphingolipid biosynthesis - lacto and neolacto series
Glycosphingolipid biosynthesis - ganglio series
Glycosphingolipid biosynthesis - globo and isoglobo series
Systemic lupus erythematosus
Type I diabetes mellitus
Allograft rejection
Graft-versus-host disease
Antigen processing and presentation
Autoimmune thyroid disease
Primary immunodeficiency
Intestinal immune network for IgA production
Asthma
Viral protein interaction with cytokine and cytokine receptor
Staphylococcus aureus infection
ER (IHC)
Positive
Negative
PR (IHC)
Positive
Negative
Indeterminate
HER2
Positive
Negative
IHC subtype
ER+/HER2-
ER+/HER2+
ER-/HER2-
ER-/HER2+
Not classified
Row z-score
≤ -2.5
0
≥ 2.5